

Self-Shape-Sensing Device with Flexible Mechanical Axes for Deformable Input Interface

Hideki Mori
Sony Interactive Entertainment, Japan
Hideki.00.Mori@sony.com

Yoshihisa Takahashi
Sony Interactive Entertainment, Japan
Yoshihisa.Takahashi@sony.com

Koichi Shiono
Sony Interactive Entertainment, Japan
Koichi.Shiono@sony.com

Hirofumi Kaneko
Sony Interactive Entertainment, Japan
Hirofumi.Kaneko@sony.com

Hiroya Matsugami
Sony Interactive Entertainment, Japan
Hiroya.Matsugami@sony.com

Masaomi Nishidate
Sony Interactive Entertainment, Japan
Masaomi.Nishidate@sony.com

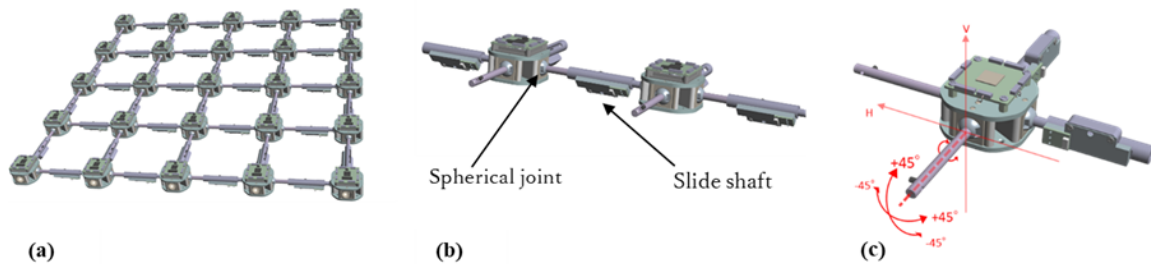


Figure 1: (a) Device overview. (b) Movable components of device. (c) Spherical joint.

ABSTRACT

We propose a novel device that is capable of sensing its own shape, structured around a flexible mechanical axis that allows for its deformation within a wider degree of freedom and enables its tangible and intuitive control by hand. This device has a 5 x 5 axis structure consisting of a spherical joint that rotates between $\pm 45^\circ$ and a slide shaft whose length can be varied from 50 to 71 mm. We developed a signal-processing algorithm which can work on the micro controller unit (MCU) of the device and use built-in sensors to not only reconstruct deformations in real-time, but also directly use such reconstructions in 3D graphics applications. As a result, we achieved an intuitive and interactive 3D experience using volumetric displays.

CCS CONCEPTS

• Human-centered computing; • Human-computer interaction (HCI); • Interaction devices;

KEYWORDS

Virtual Reality, Augmented Reality, Shape-Sensing, Tangible Control, Deformable Device

ACM Reference Format:

Hideki Mori, Yoshihisa Takahashi, Koichi Shiono, Hirofumi Kaneko, Hiroya Matsugami, and Masaomi Nishidate. 2021. Self-Shape-Sensing Device with Flexible Mechanical Axes for Deformable Input Interface. In *SIGGRAPH Asia 2021 Emerging Technologies (SA '21 Emerging Technologies)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476122.3484839>

1 INTRODUCTION

The use of interactive applications with volumetric graphics is becoming increasingly common in everyday life. This has led to the introduction of improved output devices such as XR head-mounted displays and volumetric displays in the market. A tangible interface that enables an intuitive interaction by user's hand could expand the proliferation of interactive entertainment applications with volumetric image whose control is easier for users to understand.

Previous works in this field have suggested the use of a deformable input device for intuitive computer interfaces. Such devices use a single shape-sensing strip and have been proposed in interactive applications [Chien et al., 2015]. A kind of sheet type device which can be curved has been proposed as a self-sensing input interface for 2.5D interaction [Rendl et al., 2014]. For the sensing of complex deformations, an approach has been proposed based on lattice structures involving a bendable joint [HOSHI and SHINODA, 2008].

The aforementioned examples focus on sensing deformations made up of bends and curves and utilizing such information. In this research, we propose a novel, self-sensing device which can simultaneously handle two deformation factors — bending and stretching — to enable a wider degree of freedom for the detection of plane deformation.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '21 Emerging Technologies, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8685-2/21/12.

<https://doi.org/10.1145/3476122.3484839>

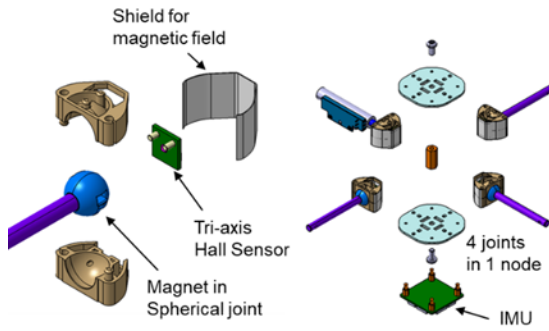


Figure 2: Spherical joint structure in each node.



Figure 3: Result of the shape-sensing activity.

2 SELF-SHAPE-SENSING DEVICE

2.1 Hardware Configuration

Figure 1(a) shows an overview of the device. Its fundamental design attributes include a 5 x 5 axis consisting of 25 nodes and 40 shafts. Figure 1(b) displays a close-up view of the movable components of each node-and-shaft pairing. The shafts can be slid to vary their lengths from 50 to 71 mm. Both ends of each shaft are attached to spherical joints that can be rotated between $\pm 45^\circ$ in all directions, as indicated in Figure 1(c).

To sense changes in the orientation of the spherical joint, we install a tri-axis Hall sensor to measure the tilt of the magnet installed into the spherical joint. Inertial measurement unit (IMU) sensors are then installed on each node to measure the tilt of the node itself (Figure 2). A slide-position sensor is placed on the shaft to detect changes in its length.

2.2 Operation

The proposed device obtains values from built-in sensors. The algorithm used on the MCU can reconstruct entire shape of device from those values in real-time at the prevailing video rate. As a result, various aspects of plane deformation such as bending, expansion, and contraction are reflected in real time in the form of 3D shape-deformation data (Figure 3).

3 APPLICATIONS

To demonstrate examples of how this device could be used, we implemented it in three interactive applications with deformable inputs.

3.1 Landscape Deformation

In this application, a reconstructed, spline-interpolated plane was used to depict the contours of a landscape. When the user deformed the landscape by hand, elements of the landscape such as plants,

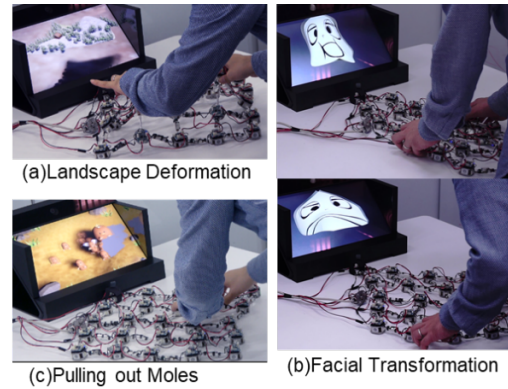


Figure 4: Example applications of the self-shape-sensing device.

lake water, and snow on the ground were changed in the output based on their relative heights in the field (Figure 4(a)).

3.2 Facial Transformation

In this application, the initial position of the nodes constituted the model of a face and relative differences from this initial layout were applied to move the characteristic points in the face. As a result, users could change the face model from its original depiction in tangible ways such as by pulling the cheek, pinching the nose, and the changing facial contours (Figure 4(b)).

3.3 Pulling Out Moles

Moles appeared at locations corresponding to the positions of nodes in the proposed device. Users could interact with these moles by moving the nodes. The application used the speed of movement of each node for pulling out a mole and the change of node density to combine multiple individual moles into a giant mole (Figure 4(c)).

4 FUTURE WORK

In our future work, we plan to explore the scalability of this device by increasing the number of axes and downsizing each part for more flexibility, a wider degree of freedom, and higher resolution. We need to consider whole node topology to balance node counts and data throughput. We will also improve the accuracy of the shape-sensing mechanism and plane reconstruction algorithm by effectively compensating the error value from each sensor.

REFERENCES

- Csohin-yu Chien, Rong-Hao Liang, Long-Fei Lin, Liwei Chan, Bing-Yu Chen. 2015. FlexiBend: Enabling Interactivity of Multi-Part, Deformable Fabrications Using Single Shape-Sensing Strip. In *Proc. ACM UIST '15* (2015), 659–663.
- Christian Rendl, David Kim, Sean Fanello, Patrick Parzer, Christoph Rhemann, Jonathan Taylor, Martin Zirkel, Gregor Scheipl, Thomas Rothlander, Michael Haller, Shahram Izadi. 2014. Flexsense: A transparent self-sensing deformable surface. In *Proc. ACM UIST '14* (2014), 129–138.
- Takayuki HOSHI, Hiroyuki SHINODA. 2008. Cloth-like Device Measuring 3D Shape in Real Time In *Proc. Transactions of the Society of Instrument and Control Engineers* 2008 Vol. 44, Issue 12, 1015–1016.